A drug-by-mutation map for childhood genetic epilepsy

The same anti-seizure drug helps one child and harms another. Nobody has measured which, for 94% of these mutations.

Vaishnavi Moturi · Harvard University · 2026

The same drug, the opposite effect

Direction of effect decides treatment, and it is invisible in a genetic report

a sodium-channel blocker such as carbamazepine

SCN1A

loss of function

harms the child

SCN8A

gain of function

first-line therapy

- Both children carry one missense mutation in one channel gene.
- Which way the channel failed decides whether the drug is therapy or injury.
- The genetic report names the mutation. It does not say the direction.
- So direction is inferred from the age seizures began, and the child goes through sequential drug trials of weeks each.

Opposite drug response by direction in the same channel family: PMID 35037686 (Brain, 2022).

Direction has essentially never been measured

Across six disease families where direction decides treatment

939

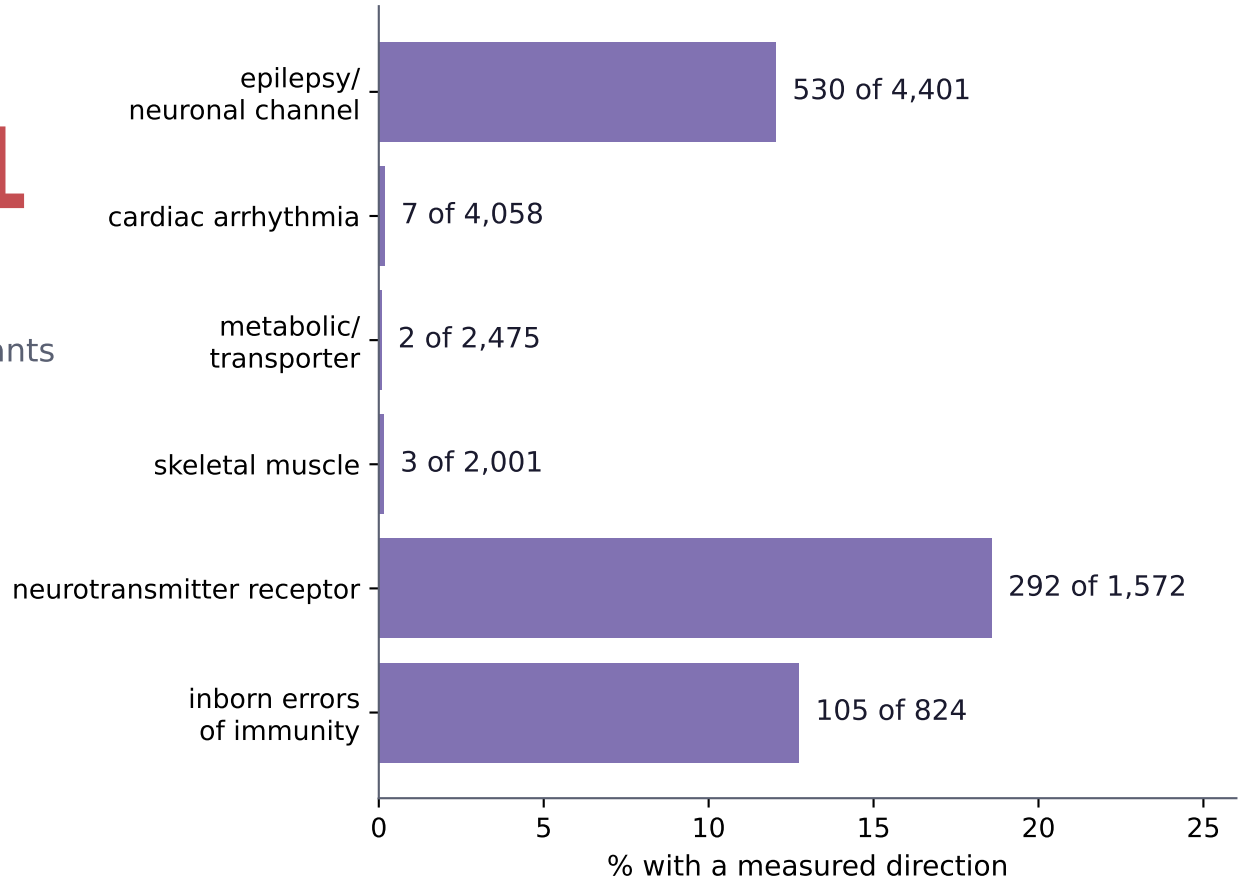
variants with a measured direction, worldwide

15,331

pathogenic missense variants that need one

6.1%

and two thirds of the 939 come from just two studies. In cardiac arrhythmia genes it is 7 of 4,058.

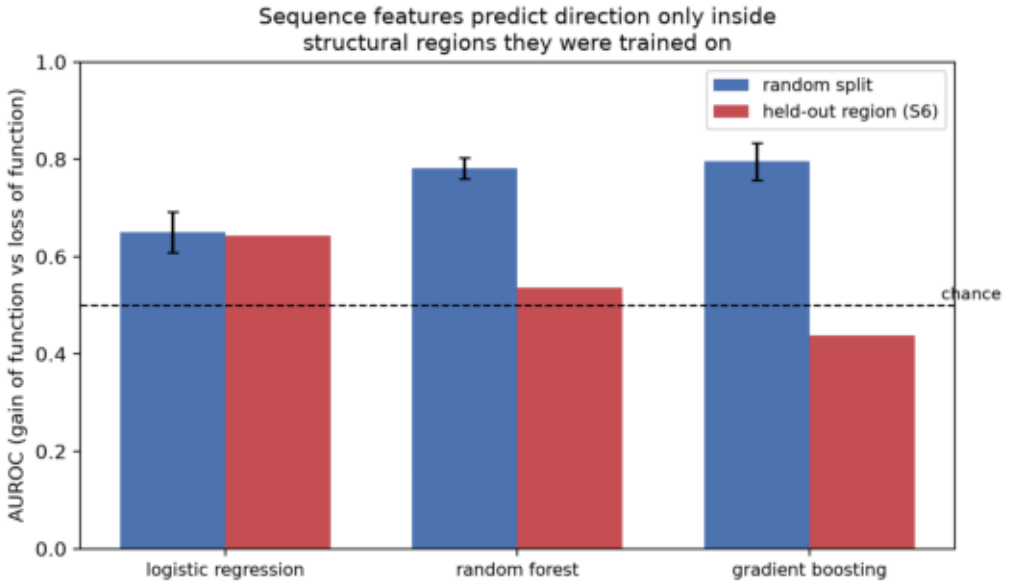


Census of 36 genes across six families; non-epilepsy rows are lower bounds. See docs/METHODS.md.

This is a data problem, not an algorithm problem

I built the model my own idea depended on, and it failed out of sample

- Trained on the largest measured set that exists: 291 KCNQ2 variants against 57 benign controls.
- Random split: AUROC 0.65 to 0.80. Looks publishable.
- Hold out one structural region, which is the real situation because a new patient's mutation is often somewhere nobody measured, and it falls to 0.44 at worst and 0.64 at best.
- The same class of method reaches AUC 0.94 in kinases, where two decades of oncology produced thousands of labels.



The difference between 0.94 and chance is the training data.

Cystic fibrosis is the precedent, not the plan

What it proves, and what it leaves unproven

PROVEN

Matching drug to mutation class is clinically real and commercially large.

Done at 655-variant scale in CFTR, one variant per well.

Underwrites the most successful rare-disease franchise in medicine.

UNPROVEN

Pooling the variants instead of arraying them, never shown at scale for an ion channel.

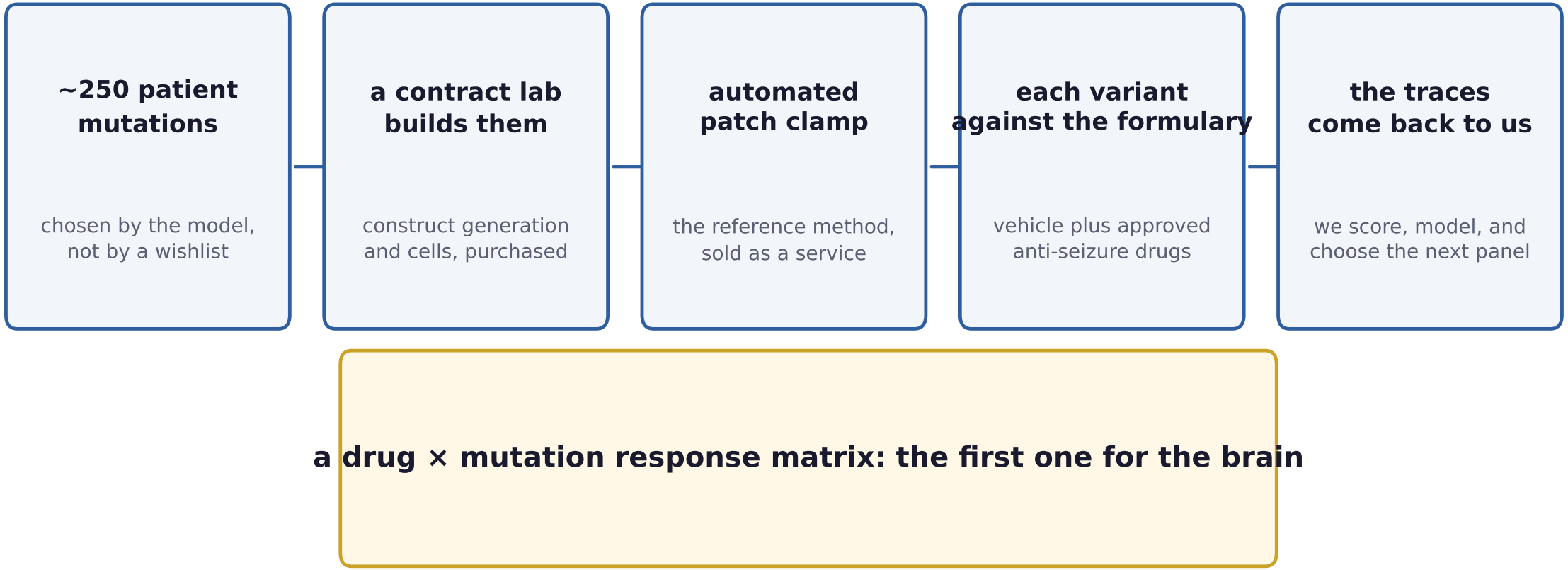
A two-sided selection: CFTR only ever breaks one way, so CF never needed one.

Transfer to another gene. CF produced a lookup table, not a model.

The unproven column is the first experiment, not an assumption. PMID 38388235 (655-variant CFTR panel).

What I would build

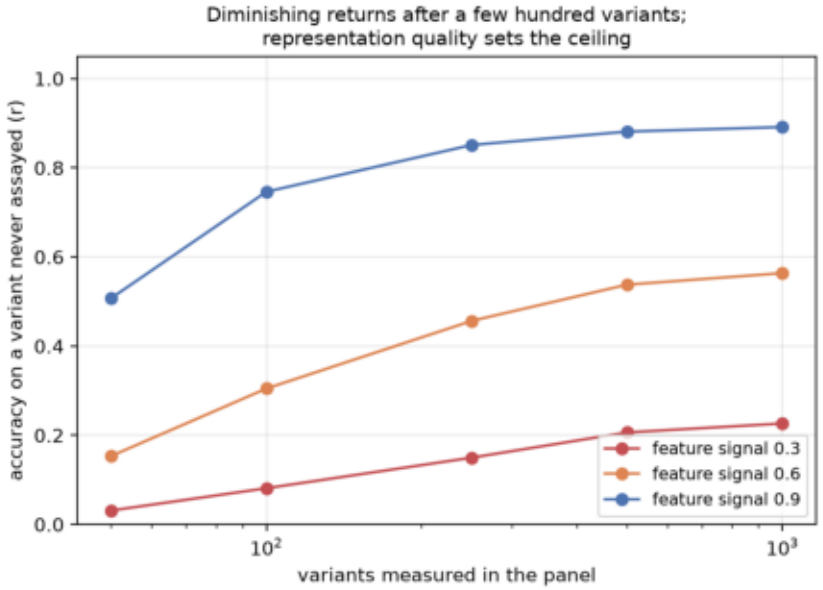
Commissioned measurement, not a laboratory of my own

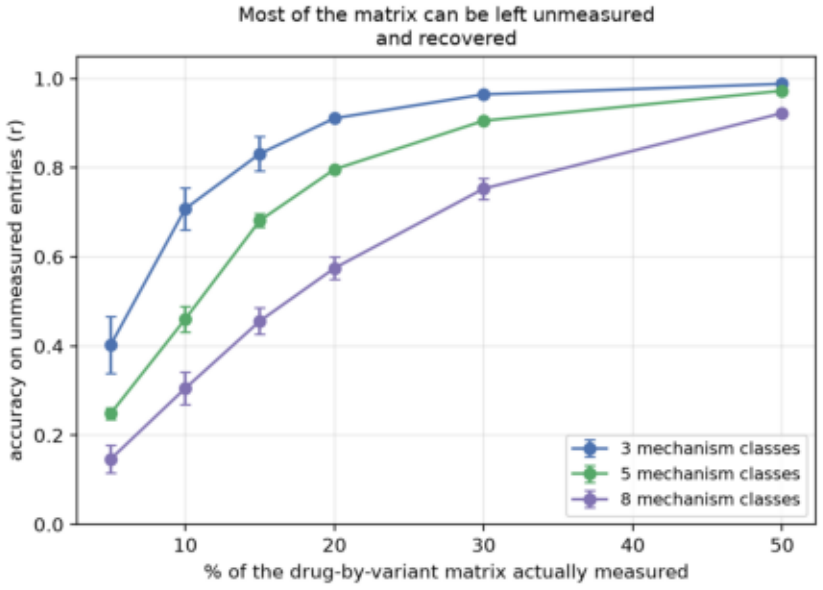


No instruments, no facility, no pipette of mine. Pooling the variants into one culture is the later cost-reduction step, once panels run to thousands.

The first experiment is small on purpose

Simulation sized it before any money was spent





250

variants is where the returns flatten, so that is the panel

- Growing the panel from 50 to 250 adds about 0.3 to the correlation. From 250 to 1,000 adds 0.04 to 0.11.
- Most of the matrix can be left unmeasured and recovered: r = 0.80 from 20% of entries.
- So spend on drug conditions and better representations, not on more mutations.

What the model does with it

Four moves, in the order they become possible

1

factorise

the matrix's latent dimensions ARE the mechanism classes: discovered, not curated

2

complete

fill the entries nobody measured; 20 to 30% coverage recovers the rest

3

cold start

predict a never-assayed variant's whole drug profile from its sequence

4

transfer

share structure across paralogous channels with gene-specific offsets

Cystic fibrosis can say nothing about variant 656. The point of building the map is to stop needing it.

Move 4 is the one that decides platform versus dataset, and it needs measured variants in a second gene.

Who pays, and for what

The report already exists; the direction column does not

MEASURED	in our panel, with assay-vs-patch-clamp concordance stated
PREDICTED	not in the panel; model call with held-out calibration attached
UNDETERMINED	we decline to call it: the tier that keeps the product honest

- Sold to the diagnostic laboratories that already return these genetic reports. Not to hospitals, and not to patients.
- Second buyer: the handful of companies running direction-dependent epilepsy programmes, for trial enrichment.
- We report the biology. The clinician prescribes. That line is the regulatory position and we do not cross it.

Where it goes after epilepsy

Three steps, each with its number attached

neurodevelopmental conditions

The same genes cause autism and intellectual disability. In several, the direction decides which one a child develops.

29 of 142

epilepsy genes also in a 492-gene neurodevelopmental set

0 of 91 industry autism trials used a physiological primary endpoint

other channel diseases

The same architecture carries disease in the heart, skeletal muscle and the inner ear, and is just as unmeasured.

148

human ion channel genes carry disease-causing mutations

7 of 4,058 measured in cardiac arrhythmia genes

the ceiling, stated as a ceiling

Any protein where a cell can be made to die when it does too much, and another when it does too little.

2,503 of 3,994

human disease genes in such a class

each needs its own assay built, and some will fail: a reach, not a roadmap

The drug does not transfer across these; the measurement does.

What is true today

Stated plainly, because the gaps are the plan

built

- ✓ 142-gene landscape, mechanism and competition mapped
- ✓ the 939-of-15,331 measurement census
- ✓ a benchmark of my own that failed out of sample
- ✓ a simulation that sized and cheapened the first experiment
- ✓ every number reproducible from a public repository

absent

- ✗ any experiment of my own
- ✗ a cost per variant: no vendor quote has been obtained
- ✗ an assay-design co-founder who can run the rig
- ✗ proof that pooling works on a channel at scale
- ✗ an entity, a team, or assignable IP

The ask

Four things, none of which needs a lab of my own

- An answer from the two groups who published the pooled assay in 2020 and stopped after twelve amino acids: technical wall, or change of priorities?
- Three vendor quotes, turning the first screen from an estimate into a budget.
- A collaborating group that can run it, and a baseline model trained on all public data so the number to beat is one I generated.
- The transfer test in a second sodium channel, the experiment that decides whether this is a platform or one gene at a time.

Cystic fibrosis measured variants in order to sell a drug.

I want to measure variants in order to stop having to measure them.